

# Annex E – Data reported on MRSA from food-producing animals and derived meat

## Annex to:

EFSA (European Food Safety Authority) and ECDC (European Centre for Disease Prevention and Control), 2025. The European Union Summary Report on Antimicrobial Resistance in zoonotic and indicator bacteria from humans, animals and food in 2023-2024.

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## E.1 Methicillin-resistant *Staphylococcus aureus* in food

**Table 1: Methicillin-resistant *Staphylococcus aureus* in food, 2024.**

| Food                                        | Country                     | Production type/monitoring description (where specified)                                | Sample unit | Units tested | Units positive for MRSA (%) |
|---------------------------------------------|-----------------------------|-----------------------------------------------------------------------------------------|-------------|--------------|-----------------------------|
| Meat from broilers ( <i>Gallus gallus</i> ) | Austria                     | Meat from broilers ( <i>Gallus gallus</i> ) - fresh - chilled                           | batch       | 325          | 15 (4.6%)                   |
|                                             | Italy                       | Meat from broilers ( <i>Gallus gallus</i> ) - fresh - chilled                           | single      | 2            | 2 (100%)                    |
|                                             | Norway                      | Meat from broilers ( <i>Gallus gallus</i> ) - fresh                                     | batch       | 325          | 0 (0%)                      |
|                                             | Republic of North Macedonia | Meat from broilers ( <i>Gallus gallus</i> ) - fresh - chilled                           | single      | 4            | 0 (0%)                      |
|                                             | Netherlands                 | Meat from broilers ( <i>Gallus gallus</i> ) - fresh - chilled                           | single      | 407          | 28 (6.9%)                   |
|                                             | Germany                     | Meat from broilers ( <i>Gallus gallus</i> ) - carcass                                   | single      | 405          | 110 (27.2%)                 |
|                                             | Germany                     | Meat from broilers ( <i>Gallus gallus</i> ) - fresh - chilled - border control post     | single      | 26           | 0 (0%)                      |
|                                             | Germany                     | Meat from broilers ( <i>Gallus gallus</i> ) - fresh - chilled - organic production      | single      | 286          | 7 (2.4%)                    |
|                                             | Germany                     | Meat from broilers ( <i>Gallus gallus</i> ) - fresh - chilled - conventional production | single      | 386          | 23 (6%)                     |
| Meat from turkey                            | Austria                     | Meat from turkey - fresh - chilled                                                      | batch       | 173          | 28 (16.2%)                  |

| Food                        | Country                     | Production type/monitoring description (where specified)         | Sample unit | Units tested | Units positive for MRSA (%) |
|-----------------------------|-----------------------------|------------------------------------------------------------------|-------------|--------------|-----------------------------|
|                             | Italy                       | Meat from turkey - fresh - chilled                               | single      | 1            | 1 (100%)                    |
|                             | Norway                      | Meat from turkey - fresh                                         | batch       | 114          | 0 (0%)                      |
|                             | Netherlands                 | Meat from turkey - fresh - chilled                               | single      | 21           | 5 (23.8%)                   |
|                             | Germany                     | Meat from turkey - carcase                                       | single      | 400          | 226 (56.5%)                 |
|                             | Germany                     | Meat from turkey - fresh - chilled                               | single      | 355          | 128 (36,1%)                 |
| Meat from bovine animals    | Republic of North Macedonia | Meat from bovine animals - fresh - chilled                       | single      | 4            | 0 (0%)                      |
|                             | Netherlands                 | Meat from bovine animals - fresh                                 | single      | 475          | 36 (7.6%)                   |
| Meat from pig               | Republic of North Macedonia | Meat from pig - fresh - chilled                                  | single      | 7            | 0 (0%)                      |
|                             | Netherlands                 | Meat from pig - fresh                                            | single      | 279          | 22 (7.9%)                   |
| Meat from sheep             | Netherlands                 | Meat from sheep - fresh                                          | single      | 124          | 6 (4.8%)                    |
|                             | Netherlands                 | Meat from sheep - meat preparation - intended to be eaten cooked | single      | 30           | 1 (3.3%)                    |
| Crustaceans                 | Germany                     | Crustaceans - unspecified - raw                                  | single      | 226          | 40 (17.7%)                  |
| Meat from deer (venison)    | Netherlands                 | Meat from deer (venison) - fresh - frozen                        | batch       | 18           | 0 (0%)                      |
| Meat from wild game - birds | Netherlands                 | Meat from wild game - birds - fresh                              | batch       | 4            | 0 (0%)                      |

**Table 2: Methicillin-resistant *Staphylococcus aureus* in food, 2023.**

| Food                                        | Country     | Production type/monitoring description (where specified) | Sample unit | Units tested | Positive for MRSA (%)     |
|---------------------------------------------|-------------|----------------------------------------------------------|-------------|--------------|---------------------------|
| Meat from bovine animals                    | Austria     | Fresh - retail monitoring                                | Single      | 310          | 6 (1.9%) <sup>(a)</sup>   |
|                                             | Germany     | Fresh - sampling at border control posts (monitoring)    | Single      | 74           | 5 (6.8%) <sup>(b)</sup>   |
|                                             | Netherlands | Fresh - processing plant monitoring                      | Batch       | 184          | 25 (13.6%) *              |
|                                             | Netherlands | Fresh - retail monitoring                                | Single      | 432          | 17 (3.9%) *               |
| Meat from broilers ( <i>Gallus gallus</i> ) | Netherlands | Fresh - retail monitoring                                | Single      | 185          | 18 (9.7%) *               |
| Meat from pigs                              | Austria     | Fresh - retail monitoring                                | Single      | 313          | 45 (14.4%) <sup>(c)</sup> |
|                                             | Germany     | Fresh - retail monitoring                                | Single      | 380          | 39 (10.3%) <sup>(d)</sup> |
|                                             | Netherlands | Fresh - retail monitoring                                | Single      | 291          | 20 (6.9%) *               |
|                                             | Slovakia    | Fresh - retail monitoring                                | Single      | 130          | 20 (15.4%) *              |
| Meat from turkey                            | Netherlands | Fresh - retail monitoring                                | Single      | 7            | 2 (28.6%) *               |
| Meat from deer (venison)                    | Netherlands | Fresh - sampling at border control posts (monitoring)    | Batch       | 11           | 0 (0%)                    |
| Meat from duck                              | Netherlands | Fresh - sampling at border control posts (monitoring)    | Batch       | 3            | 0 (0%)                    |
| Meat from farmed game - land mammals        | Netherlands | Fresh - sampling at border control posts (monitoring)    | Batch       | 3            | 0 (0%)                    |
| Meat from sheep                             | Netherlands | Fresh - retail monitoring                                | Single      | 151          | 7 (4.6%) *                |

|                             |             |                                                       |        |    |             |
|-----------------------------|-------------|-------------------------------------------------------|--------|----|-------------|
| Meat from wild game - birds | Netherlands | Fresh - sampling at border control posts (monitoring) | Batch  | 5  | 1 (20.0%) * |
| Prawns                      | Germany     | Fresh - primary production monitoring                 | Single | 10 | 0 (0%)      |

<sup>(a)</sup> *spa*-types: t011 (3), t034 (3)

<sup>(b)</sup> *spa*-types: t002 (4), t008 (1)

<sup>(c)</sup> *spa*-types: t011 (20), t034 (12), t899 (1), t1255 (1), t1430 (3), t1451 (2), t2011 (4), t5524 (1), CC398, ST398 *spa*-type not reported (1)

<sup>(d)</sup> *spa*-types: t008 (1), t011 (10), t034 (7), t127 (1), t693 (1), t899 (8), t1419 (1), t1430 (2), t1451 (1), t1580 (1), t2011 (2), t3512 (1), t20072 (1), *spa*-type not reported (2)

\**spa*-types not reported

## E.2 Methicillin-resistant *Staphylococcus aureus* in food-producing animals, clinical investigations excluded

**Table 3: Methicillin-resistant *Staphylococcus aureus* in targeted food-producing animals, clinical investigations excluded, 2024.**

| Animal                      | Country                     | Production type                                                          | Sample unit     | Units tested | Positive for MRSA (%) |
|-----------------------------|-----------------------------|--------------------------------------------------------------------------|-----------------|--------------|-----------------------|
| Cattle (bovine animals)     | Belgium                     | Cattle (bovine animals) - calves (under 1 year) - veal calves            | herd/flock      | 145          | 64 (44.1%)            |
|                             | Belgium                     | Cattle (bovine animals) - dairy cows                                     | herd/flock      | 126          | 12 (9.5%)             |
|                             | Belgium                     | Cattle (bovine animals) - meat production animals                        | herd/flock      | 85           | 8 (9.4%)              |
|                             | Netherlands                 | Cattle (bovine animals) - dairy cows                                     | animal          | 674          | 0 (0%)                |
|                             | Republic of North Macedonia | Cattle (bovine animals) - calves (under 1 year)                          | slaughter batch | 13           | 0 (0%)                |
|                             | Republic of North Macedonia | Cattle (bovine animals) - dairy cows                                     | herd/flock      | 38           | 0 (0%)                |
| <i>Gallus gallus</i> (fowl) | Germany                     | <i>Gallus gallus</i> (fowl) - broilers - before slaughter - conventional | single          | 261          | 3 (1.1%)              |
|                             | Germany                     | <i>Gallus gallus</i> (fowl) - broilers - before slaughter - organic      | single          | 61           | 0 (0%)                |
|                             | Norway                      | <i>Gallus gallus</i> (fowl)                                              | animal          | 325          | 0 (0%)                |
|                             | Republic of North Macedonia | <i>Gallus gallus</i> (fowl) - broilers - before slaughter                | herd/flock      | 4            | 0 (0%)                |
| Pigs                        | Netherlands                 | Pigs - fattening pigs                                                    | holding         | 114          | 81 (71.1%)            |
|                             | Norway                      | Pigs - breeding animals                                                  | herd/flock      | 220          | 0 (0%)                |
|                             | Norway                      | Pigs - fattening pigs                                                    | animal          | 27           | 0 (0%)                |



|                |                             |                            |                 |      |           |
|----------------|-----------------------------|----------------------------|-----------------|------|-----------|
|                | Norway                      | Pigs - fattening pigs      | herd/flock      | 2449 | 0 (0%)    |
|                | Republic of North Macedonia | Pigs - fattening pigs      | herd/flock      | 24   | 0 (0%)    |
|                | Republic of North Macedonia | Pigs - fattening pigs      | slaughter batch | 12   | 0 (0%)    |
| Turkeys        | Norway                      | Turkeys - fattening flocks | animal          | 113  | 0 (0%)    |
| Small ruminant | Portugal <sup>‡</sup>       | Goats                      | herd/flock      | 34   | 4 (11.7%) |
|                | Portugal <sup>‡</sup>       | Sheep                      | herd/flock      | 72   | 4 (5.4%)  |

<sup>‡</sup>Data retrieved from antimicrobial resistance isolate based data model.



**Table 4: Methicillin-resistant *Staphylococcus aureus* in targeted food-producing animals, clinical investigations excluded, 2023.**

| Animal                      | Country     | Production type                                                | Sample unit | Units tested | Positive for MRSA (%) |
|-----------------------------|-------------|----------------------------------------------------------------|-------------|--------------|-----------------------|
| Pigs                        | Norway      | Pigs                                                           | herd/flock  | 541          | 0 (0%)                |
|                             | Switzerland | Pigs - fattening pigs                                          | animal      | 310          | 166 (53.5%)           |
|                             | Slovakia    | Pigs                                                           | animal      | 130          | 27 (20.8%)            |
|                             | Germany     | Pigs - breeding animals                                        | herd/flock  | 351          | 91 (25.9%)            |
| Cattle (bovine animals)     | Netherlands | Cattle (bovine animals) - dairy cows                           | animal      | 316          | 2 (0.6%)              |
|                             | Switzerland | Cattle (bovine animals) - calves (under 1 year)                | animal      | 307          | 11 (3.6%)             |
| <i>Gallus gallus</i> (fowl) | Belgium     | <i>Gallus gallus</i> (fowl) - laying hens - adult              | herd/flock  | 33           | 0 (0%)                |
|                             | Belgium     | <i>Gallus gallus</i> (fowl) - broilers - during rearing period | herd/flock  | 96           | 1 (1%)                |
| Small ruminants             | Netherlands | Sheep - meat production animals                                | herd/flock  | 156          | 8 (5.1%)              |
| Turkeys                     | Belgium     | Turkeys - fattening flocks                                     | herd/flock  | 27           | 5 (18.5%)             |

### E.3 Methicillin-resistant *Staphylococcus aureus* in food-producing animals, clinical investigations

**Table 5: Methicillin-resistant *Staphylococcus aureus* in targeted food-producing animals, clinical investigations, 2024.**

| Animal          | Country     | Production type                                  | Sample unit | Units tested | Positive for MRSA (%) |
|-----------------|-------------|--------------------------------------------------|-------------|--------------|-----------------------|
| Cattle          | Norway      | Cattle (bovine animals) – outbreak investigation | animal      | 69           | 13 (18.8%)            |
|                 | Norway      | Cattle (bovine animals) – outbreak investigation | herd/flock  | 69           | 27 (39.1%)            |
| Pigs            | Italy       | Pigs                                             | animal      | 1            | 1 (100%)              |
|                 | Netherlands | Pigs - unspecified                               | animal      | 20           | 2 (10%)               |
|                 | Norway      | Pigs - mixed herds                               | animal      | 3            | 0 (0%)                |
|                 | Norway      | Pigs - mixed herds                               | herd/flock  | 2            | 0 (0%)                |
| Small ruminants | Netherlands | Goats                                            | animal      | 7            | 1 (14.3%)             |

## E.4 Methicillin-resistant *Staphylococcus aureus* in non-food-producing animals, clinical investigations

**Table 66: Methicillin-resistant *Staphylococcus aureus* in targeted non-food-producing animals, clinical investigations, 2024.**

| Animal             | Country     | Production type                 | Sample unit | Units tested | Positive for MRSA (%) |
|--------------------|-------------|---------------------------------|-------------|--------------|-----------------------|
| Dogs               | Norway      | Dogs - pet animals              | animal      | 4            | 0 (0%)                |
|                    | Netherlands | Dogs - pet animals              | animal      | 7447         | 10 (0.1%)             |
| Felidae            | Netherlands | Cats - pet animals              | animal      | 2749         | 5 (0.2%)              |
|                    | Norway      | Cats - pet animals              | Animal      | 1            | 0 (0%)                |
| Land game mammals  | Netherlands | Land game mammals - zoo animals | animal      | 470          | 16 (3.4%)             |
| Solipeds, domestic | Netherlands | Solipeds, domestic - horses     | animal      | 928          | 15 (1.6%)             |

**Table 77: Methicillin-resistant *Staphylococcus aureus* in targeted non-food-producing animals, clinical investigations, 2023.**

| Animal             | Country     | Production type             | Sample unit | Units tested | Positive for MRSA (%) |
|--------------------|-------------|-----------------------------|-------------|--------------|-----------------------|
| Dogs               | Netherlands | Dogs                        | animal      | 7969         | 8 (0.1%)              |
| Felidae            | Netherlands | Cats - pet animals          | animal      | 2883         | 3 (0.1%)              |
| Solipeds, domestic | Netherlands | Solipeds, domestic - horses | animal      | 1028         | 13 (1.3%)             |

## E.5 *spa*-type characterization in methicillin-resistant *Staphylococcus aureus*

**Table 88: Methicillin-resistant *Staphylococcus aureus* *spa*-type characterization, 2024.**

| Category | Country | Animal/food type                            | Sample type/unit         | No. of isolates/<br>No. of samples tested | <i>spa</i> -type | ST   | CC  | Inferred CC* | LA, CA or HA type <sup>y</sup> | Inferred CC & type | <i>mec</i>  | PVL status/IEC genes |
|----------|---------|---------------------------------------------|--------------------------|-------------------------------------------|------------------|------|-----|--------------|--------------------------------|--------------------|-------------|----------------------|
| Food     | Austria | Meat from broilers ( <i>Gallus gallus</i> ) | Batch (food/feed)        | 5/325                                     | 34               | 398  | 398 | -            | LA                             | CC398/LA           | <i>mecA</i> | -                    |
|          |         |                                             |                          | 2/325                                     | 11               | 398  | 398 | -            | LA                             | CC398/LA           | <i>mecA</i> | -                    |
|          |         |                                             |                          | 1/325                                     | 330              | 45   | 45  | -            | -                              | CC45               | <i>mecA</i> | -                    |
|          |         |                                             |                          | 1/325                                     | 330              | 9338 | 45  | -            | -                              | CC45               | <i>mecA</i> | -                    |
|          |         |                                             |                          | 2/325                                     | 899              | 398  | 398 | -            | LA                             | CC398/LA           | <i>mecA</i> | -                    |
|          |         |                                             |                          | 1/325                                     | 127              | 1    | 1   | -            | LA                             | CC1/LA             | <i>mecA</i> | -                    |
|          |         |                                             |                          | 1/325                                     | 2                | 149  | 5   | -            | -                              | CC5                | <i>mecA</i> | -                    |
|          |         |                                             |                          | 1/325                                     | 223              | 22   | 22  | -            | -                              | CC22               | <i>mecA</i> | -                    |
|          |         |                                             |                          | 1/325                                     | 843              | 130  | -   | -            | -                              | -                  | <i>mecC</i> | -                    |
|          |         | Meat from turkey                            | Batch (food/feed)        | 13/173                                    | 11               | 398  | 398 | -            | LA                             | CC398/LA           | <i>mecA</i> | -                    |
|          |         |                                             |                          | 1/173                                     | 11               | 5672 | 398 | -            | LA                             | CC398/LA           | -           | -                    |
|          |         |                                             |                          | 6/173                                     | 34               | 398  | 398 | -            | LA                             | CC398/LA           | <i>mecA</i> | -                    |
|          |         |                                             |                          | 5/173                                     | 899              | 398  | 398 | -            | LA                             | CC398/LA           | <i>mecA</i> | -                    |
|          |         |                                             |                          | 1/173                                     | 1255             | 398  | 398 | -            | LA                             | CC398/LA           | <i>mecA</i> | -                    |
|          |         |                                             |                          | 1/173                                     | 127              | 1    | 1   | -            | LA                             | CC1/LA             | <i>mecA</i> | -                    |
|          |         |                                             |                          | 1/173                                     | 8425             | 45   | 45  | -            | -                              | CC45               | <i>mecA</i> | -                    |
|          | Germany | Meat from turkey - carcasses                | Batch - slaughter animal | 123/400                                   | 34               | -    | -   | 398          | LA                             | CC398/LA           | <i>mecA</i> | -                    |
|          |         |                                             |                          | 16/400                                    | 899              | -    | 398 | -            | LA                             | CC398/LA           | <i>mecA</i> | -                    |
|          |         |                                             |                          | 26/400                                    | 899              | -    | -   | 398          | LA                             | CC398/LA           | <i>mecA</i> | -                    |

| Category | Country | Animal/food type                    | Sample type/<br>unit | No. of<br>isolates/<br>No. of<br>samples<br>tested | spa-type | ST  | CC      | Inferred<br>CC* | LA, CA<br>or HA<br>type <sup>‡</sup> | Inferred<br>CC &<br>type | mec  | PVL<br>status/<br>IEC<br>genes |
|----------|---------|-------------------------------------|----------------------|----------------------------------------------------|----------|-----|---------|-----------------|--------------------------------------|--------------------------|------|--------------------------------|
|          |         |                                     |                      | 27/400                                             | 11       | -   | -       | 398             | LA                                   | CC398/LA                 | mecA | -                              |
|          |         |                                     |                      | 1/400                                              | 3841     | 672 | -       | -               | -                                    |                          | mecA | -                              |
|          |         |                                     |                      | 6/400                                              | 3841     | -   | -       | -               | -                                    |                          | mecA | -                              |
|          |         |                                     |                      | 4/400                                              | 1430     | -   | -       | 1               | LA                                   | CC1/LA                   | mecA | -                              |
|          |         |                                     |                      | 3/400                                              | 21298    | -   | -       | -               | -                                    |                          | mecA | -                              |
|          |         |                                     |                      | 3/400                                              | 6575     | -   | -       | 398             | LA                                   | CC398/LA                 | mecA | -                              |
|          |         |                                     |                      | 2/400                                              | 10485    | -   | -       | 398             | LA                                   | CC398/LA                 | mecA | -                              |
|          |         |                                     |                      | 2/400                                              | 8425     | -   | -       | -               | -                                    |                          | mecA | -                              |
|          |         |                                     |                      | 1/400                                              | 8588     | -   | -       | -               | -                                    |                          | mecA | -                              |
|          |         |                                     |                      | 1/400                                              | 8588     | 45  | -       | -               | -                                    |                          | mecA | -                              |
|          |         |                                     |                      | 1/400                                              | 1255     | -   | -       | 398             | LA                                   | CC398/LA                 | mecA | -                              |
|          |         |                                     |                      | 1/400                                              | 1422     | -   | -       | 692             | LA                                   | CC692/LA                 | mecA | -                              |
|          |         |                                     |                      | 1/400                                              | 1451     | -   | -       | 398             | LA                                   | CC398/LA                 | mecA | -                              |
|          |         |                                     |                      | 1/400                                              | 693      | -   | -       | -               | LA                                   |                          | mecA | -                              |
|          |         | Meat from<br>turkey –<br>fresh meat | Batch<br>(food/feed) | 56/355                                             | 34       | -   | -       | 398             | LA                                   | CC398/LA                 | mecA | -                              |
|          |         |                                     |                      | 13/355                                             | 899      | -   | 398     | -               | LA                                   | CC398/LA                 | mecA | -                              |
|          |         |                                     |                      | 8/355                                              | 899      | -   | Non-398 | 1/398           | LA                                   | CC1/CC398 / LA           | mecA | -                              |
|          |         |                                     |                      | 15/355                                             | 11       | -   | -       | 398             | LA                                   | CC398/LA                 | mecA | -                              |
|          |         |                                     |                      | 3/355                                              | 1451     | -   | -       | 398             | LA                                   | CC398/LA                 | mecA | -                              |
|          |         |                                     |                      | 1/355                                              | 8425     | 45  | 45      | -               | -                                    | CC45                     | mecA | -                              |
|          |         |                                     |                      | 1/355                                              | 8425     | -   | -       | -               | -                                    |                          | mecA | -                              |
|          |         |                                     |                      | 1/355                                              | 10204    | -   | -       | 1               | LA                                   | CC1/LA                   | mecA | -                              |

| Category | Country | Animal/food type                                                     | Sample type/<br>unit                   | No. of<br>isolates/<br>No. of<br>samples<br>tested | spa-type | ST  | CC  | Inferred<br>CC* | LA, CA<br>or HA<br>type <sup>‡</sup> | Inferred<br>CC &<br>type | mec  | PVL<br>status/<br>IEC<br>genes |
|----------|---------|----------------------------------------------------------------------|----------------------------------------|----------------------------------------------------|----------|-----|-----|-----------------|--------------------------------------|--------------------------|------|--------------------------------|
|          |         |                                                                      |                                        | 1/355                                              | 127      | -   | -   | 1               | LA                                   | CC1/LA                   | mecA | -                              |
|          |         |                                                                      |                                        | 1/355                                              | 1430     | -   | -   | 1               | LA                                   | CC1/LA                   | mecA | -                              |
|          |         |                                                                      |                                        | 1/355                                              | 2        | -   | -   | 5               | HA                                   | CC5/HA                   | mecA | -                              |
|          |         |                                                                      |                                        | 1/355                                              | 3841     | -   | -   | -               | -                                    |                          | mecA | -                              |
|          |         |                                                                      |                                        | 1/355                                              | 8588     | -   | -   | -               | -                                    |                          | mecA | -                              |
|          |         | Meat from<br>broilers<br>( <i>Gallus<br/>gallus</i> )-<br>carcasses  | Batch -<br>slaughter<br>animal         | 66/405                                             | 34       | -   | -   | 398             | LA                                   | CC398/LA                 | mecA | -                              |
|          |         |                                                                      |                                        | 28/405                                             | 11       | -   | -   | 398             | LA                                   | CC398/LA                 | mecA | -                              |
|          |         |                                                                      |                                        | 2/405                                              | 1344     | -   | -   | -               | -                                    |                          | mecA | -                              |
|          |         |                                                                      |                                        | 2/405                                              | 899      | -   | -   | 398             | LA                                   | CC398/LA                 | mecA | -                              |
|          |         |                                                                      |                                        | 1/405                                              | 12314    | 398 | 398 | -               | -                                    | CC398/LA                 | mecA | -                              |
|          |         |                                                                      |                                        | 1/405                                              | 1606     | -   | -   | -               | -                                    |                          | mecA | -                              |
|          |         |                                                                      |                                        | 1/405                                              | 21298    | -   | -   | -               | -                                    |                          | mecA | -                              |
|          |         |                                                                      |                                        | 1/405                                              | 4132     | -   | -   | -               | -                                    |                          | mecA | -                              |
|          |         |                                                                      |                                        | 1/405                                              | 779      | -   | -   | -               | LA                                   |                          | mecA | -                              |
|          |         | Meat from<br>broilers<br>( <i>Gallus<br/>gallus</i> )-<br>fresh meat | Batch<br>(food/feed) -<br>conventional | 1/286                                              | 6606     | -   | -   | -               | -                                    |                          | mecA | -                              |
|          |         |                                                                      |                                        | 1/286                                              | 11       | -   | -   | 398             | LA                                   | CC398/LA                 | mecA | -                              |
|          |         |                                                                      |                                        | 2/286                                              | 843      | -   | -   | -               | -                                    |                          | mecC | -                              |
|          |         |                                                                      |                                        | 2/286                                              | 899      | -   | 398 | -               | LA                                   | CC398/LA                 | mecA | -                              |
|          |         |                                                                      | Batch<br>(food/feed) -<br>organic      | 1/386                                              | 2        | -   | -   | 5               | HA                                   | CC5/HA                   | mecA | -                              |
|          |         |                                                                      |                                        | 3/386                                              | 11       | -   | -   | 398             | LA                                   | CC398/LA                 | mecA | -                              |
|          |         |                                                                      |                                        | 1/386                                              | 7        | -   | -   | -               | -                                    |                          | mecA | -                              |
|          |         |                                                                      |                                        | 1/386                                              | 12314    | -   | -   | -               | -                                    |                          | mecA | -                              |
|          |         |                                                                      |                                        | 8/386                                              | 34       | -   | -   | 398             | LA                                   | CC398/LA                 | mecA | -                              |

| Category               | Country | Animal/food type | Sample type/<br>unit | No. of<br>isolates/<br>No. of<br>samples<br>tested | spa-type | ST   | CC  | Inferred<br>CC* | LA, CA<br>or HA<br>type <sup>‡</sup> | Inferred<br>CC &<br>type | mec  | PVL<br>status/<br>IEC<br>genes |
|------------------------|---------|------------------|----------------------|----------------------------------------------------|----------|------|-----|-----------------|--------------------------------------|--------------------------|------|--------------------------------|
|                        |         | Crustaceans      | Batch<br>(food/feed) | 7/226                                              | 304      | -    | -   | -               | -                                    |                          | mecA | -                              |
|                        |         |                  |                      | 1/226                                              | 304      | 6    | -   | -               | -                                    |                          | mecA | -                              |
|                        |         |                  |                      | 5/226                                              | 127      | -    | -   | 1               | LA                                   | CC1/LA                   | mecA | -                              |
|                        |         |                  |                      | 3/226                                              | 8        | -    | -   | 8               | CA/HA                                | CC8/CA/HA                | mecA | -                              |
|                        |         |                  |                      | 2/226                                              | 189      | -    | -   | -               | -                                    |                          | mecA | -                              |
|                        |         |                  |                      | 2/226                                              | 5229     | -    | -   | -               | -                                    |                          | mecA | -                              |
|                        |         |                  |                      | 1/226                                              | 1219     | 6    | -   | -               | -                                    |                          | mecA | -                              |
|                        |         |                  |                      | 1/226                                              | 1309     | 672  | -   | -               | -                                    |                          | mecA | -                              |
|                        |         |                  |                      | 1/226                                              | 17387    | 8    | -   | -               | -                                    |                          | mecA | -                              |
|                        |         |                  |                      | 1/226                                              | 1849     | -    | -   | -               | -                                    |                          | mecA | -                              |
|                        |         |                  |                      | 1/226                                              | 19       | 722  | 30  | -               | -                                    | CC30                     | mecA | -                              |
|                        |         |                  |                      | 1/226                                              | 21860    | 1930 | -   | -               | -                                    |                          | mecA | -                              |
|                        |         |                  |                      | 1/226                                              | 21897    | 1930 | -   | -               | -                                    |                          | mecA | -                              |
|                        |         |                  |                      | 1/226                                              | 21999    | 80   | -   | -               | -                                    |                          | mecA | -                              |
|                        |         |                  |                      | 1/226                                              | 22059    | 1232 | 398 | -               | -                                    | CC398                    | mecA | -                              |
|                        |         |                  |                      | 1/226                                              | 309      | 22   | 22  | -               | -                                    | CC22                     | mecA | -                              |
|                        |         |                  |                      | 1/226                                              | 34       | -    | -   | 398             | LA                                   | CC398/LA                 | mecA | -                              |
|                        |         |                  |                      | 1/226                                              | 376      | -    | -   | -               | -                                    |                          | mecA | -                              |
|                        |         |                  |                      | 1/226                                              | 45       | -    | -   | -               | -                                    |                          | mecA | -                              |
|                        |         |                  |                      | 1/226                                              | 8588     | -    | 398 | -               | -                                    | CC398                    | mecA | -                              |
| Food-producing animals | Belgium | Calves           | Herd/flock           | 49/145                                             | 11       | 398  | 398 | -               | LA                                   | CC398/LA                 | mecA | -                              |
|                        |         |                  |                      | 1/145                                              | 11       | 6035 | 398 | -               | LA                                   | CC398/LA                 | mecA | sak, scn                       |
|                        |         |                  |                      | 1/145                                              | 1344     | 398  | 398 | -               | -                                    | CC398                    | mecA | -                              |

| Category | Country               | Animal/food type                 | Sample type/unit | No. of isolates/<br>No. of samples tested | spa-type | ST   | CC  | Inferred CC* | LA, CA or HA type <sup>‡</sup> | Inferred CC & type | mec  | PVL status/<br>IEC genes |
|----------|-----------------------|----------------------------------|------------------|-------------------------------------------|----------|------|-----|--------------|--------------------------------|--------------------|------|--------------------------|
|          |                       |                                  |                  | 2/145                                     | 1451     | 398  | 398 | -            | LA                             | CC398/LA           | mecA | -                        |
|          |                       |                                  |                  | 1/145                                     | 1606     | 398  | 398 | -            | -                              | CC398              | mecA | -                        |
|          |                       |                                  |                  | 5/145                                     | 2011     | 398  | 398 | -            | LA                             | CC398/LA           | mecA | -                        |
|          |                       |                                  |                  | 1/145                                     | 2346     | 398  | 398 | -            | LA                             | CC398/LA           | mecA | -                        |
|          |                       |                                  |                  | 3/145                                     | 2970     | 398  | 398 | -            | -                              | CC398              | mecA | -                        |
|          |                       |                                  |                  | 1/145                                     | 3423     | 398  | 398 | -            | LA                             | CC398/LA           | mecA | -                        |
|          |                       | Diary cows                       | Herd/flock       | 10/126                                    | 11       | 398  | 398 | -            | LA                             | CC398/LA           | mecA | -                        |
|          |                       |                                  |                  | 1/126                                     | 588      | 398  | 398 | -            | LA                             | CC398/LA           | mecA | -                        |
|          |                       |                                  |                  | 1/126                                     | 1456     | 398  | 398 | -            | LA                             | CC398/LA           | mecA | -                        |
|          |                       | Meat production animals (cattle) | Herd/flock       | 6/85                                      | 11       | 398  | 398 | -            | LA                             | CC398/LA           | mecA | -                        |
|          |                       |                                  |                  | 1/85                                      | 108      | 398  | 398 | -            | LA                             | CC398/LA           | mecA | -                        |
|          |                       |                                  |                  | 1/85                                      | 177      | 9381 | 1   | -            | -                              | CC1                | mecA | sak, scn                 |
|          | Germany               | Gallus gallus (fowl)             | Herd/flock       | 3/261                                     | 34       | -    | -   | 398          | LA                             | CC398/LA           | mecA | -                        |
|          | Portugal <sup>‡</sup> | Small ruminants (goats)          | Herd/flock       | 1/34                                      | 3570     | 130  | -   | -            | -                              |                    | mecC | -                        |
|          |                       |                                  |                  | 1/34                                      | 3750     | 2328 | -   | -            | -                              |                    | -    | -                        |
|          |                       |                                  |                  | 1/34                                      | 45       | 5    | -   | -            | -                              |                    | -    | -                        |
|          |                       |                                  |                  | 1/34                                      | 3        | 6048 | -   | -            | -                              |                    | -    | -                        |
|          |                       | Small ruminants (sheep)          | Herd/flock       | 1/72                                      | -        | 1640 | -   | -            | -                              |                    | -    | -                        |
|          |                       |                                  |                  | 1/72                                      | 108      | 398  | -   | -            | LA                             |                    | mecA | -                        |
|          |                       |                                  |                  | 1/72                                      | 3570     | 130  | -   | -            | -                              |                    | mecC | -                        |
|          |                       |                                  |                  | 1/72                                      | 22331    | 133  | -   | -            | -                              |                    | -    | -                        |



CC, clonal complex; ST, sequence type; CA, community-associated; HA, hospital-associated; LA, livestock-associated; IEC, immune evasion cluster; PVL, Panton-Valentin leukocidin.

Details on the categorisation of *spa*-types can be found in the chapter on MRSA in the main part of the report (Section 6.3.3 Food and animals: Results of molecular typing of isolates). A summary of the information presented in this table is also visualised in Figure 60 of the MRSA chapter in the main part of the report.

\*Data retrieved from antimicrobial resistance isolate based data model.

\*For isolates where inferred CC is stated, the CC is based on the *spa*-type or ST as reported in literature:

- If ST and/or CC was not specified, *spa*-type t899 was classified as CC1/CC398 (EFSA, 2009; Guardabassi et al., 2009; Larsen et al., 2016).
- *spa*-types t011, t034, t108, t571, t588, t1255, t1451, t1456, t1580, t1793, t2011, t2330, t2346, t2576, t2922, t5452, t6228, t6575, t10485, t19248 were classified as CC398 (Battisti et al., 2010; EFSA 2009; Kinross et al., 2017; Köck et al., 2013; Pauly et al., 2019; Tkadlec et al., 2023).
- *spa*-type t127 was classified as CC1, LA-MRSA (EFSA, 2009; Merialdi et al., 2019).
- *spa*-types t002 and t242 were classified as CC5 (Asanin et al., 2019; Köck et al., 2013).
- *spa*-types t008 and t009 were classified as CC8 (Boost et al., 2012; Cuny et al., 2016).
- *spa*-types t1419, t1430 and t10204 were associated to ST9 (EFSA, 2009; Hasman et al., 2011; Köck et al., 2013) and classified as CC1 (PubMLST<sup>1</sup>).
- *spa*-type t1422 was classified as CC692 (Silva et al., 2020).
- *spa*-type t3512 was classified as ST2343 (Chen et al., 2017) and CC1 (PubMLST<sup>1</sup>).
- ST9 and ST8325 were classified as CC1 by PubMLST<sup>1</sup>.
- ST22 was classified as CC22 by PubMLST<sup>1\*</sup>

\*Isolates with reported *spa* types were classified into their putative host association groups according to the following schedule:

- The following *spa* types were classified as: HA (Hospital associated): 2, 242, 20072
- The following *spa* types were classified as: CA/HA (Community or Hospital associated): 8, 9
- The following *spa* types were classified as: LA (Livestock associated): 11, 34, 108, 127, 235, 538, 571, 588, 693, 779, 899, 1255, 1419, 1422, 1430, 1451, 1456, 1457, 1580, 1793, 2011, 2330, 2346, 2576, 2922, 3075, 3423, 3512, 4652, 5104, 5452, 5524, 6228, 6575, 6608, 10204, 10485, 14089, 15528, 16666, 19248, 21217.

<sup>1</sup> [https://pubmlst.org/bigsdb?db=pubmlst\\_saureus\\_seqdef&page=query](https://pubmlst.org/bigsdb?db=pubmlst_saureus_seqdef&page=query)

**Table 99: Methicillin-resistant *Staphylococcus aureus* spa-type characterization, 2023.**

| Category | Country | Animal/food type | Sample type/unit   | No. of isolates | spa-type | PVL status/IEC genes | ST  | CC  | mec-gene | Inferred CC <sup>‡</sup> | LA, CA or HA | Inferred ST/CC & type |
|----------|---------|------------------|--------------------|-----------------|----------|----------------------|-----|-----|----------|--------------------------|--------------|-----------------------|
| Food     | Austria | Meat from cattle | Single, fresh, RM  | 2/310           | 11       | -                    | 398 | 398 | mecA     |                          | LA           | CC398/LA              |
|          |         |                  |                    | 3/310           | 34       | -                    | 398 | 398 | mecA     |                          | LA           | CC398/LA              |
|          |         |                  |                    | 1/310           | 11       | -                    | -   | 398 | mecA     |                          | LA           | CC398/LA              |
|          |         | Meat from pigs   | Single, fresh, RM  | 1/313           | 899      | -                    | 9   | 1   | mecA     |                          | LA           | CC1/LA                |
|          |         |                  |                    | 3/313           | 1430     | -                    | 9   | 1   | mecA     |                          | LA           | CC1/LA                |
|          |         |                  |                    | 19/313          | 11       | -                    | 398 | 398 | mecA     |                          | LA           | CC398/LA              |
|          |         |                  |                    | 12/313          | 34       | -                    | 398 | 398 | mecA     |                          | LA           | CC398/LA              |
|          |         |                  |                    | 2/313           | 1451     | -                    | 398 | 398 | mecA     |                          | LA           | CC398/LA              |
|          |         |                  |                    | 4/313           | 2011     | -                    | 398 | 398 | mecA     |                          | LA           | CC398/LA              |
|          |         |                  |                    | 1/313           | 5524     | -                    | 398 | 398 | mecA     |                          | LA           | CC398/LA              |
|          |         |                  |                    | 1/313           | -        | -                    | 398 | 398 | mecA     |                          | LA           | CC398/LA              |
|          |         |                  |                    | 1/313           | 11       | -                    | -   | 398 | mecA     |                          | LA           | CC398/LA              |
|          |         |                  |                    | 1/313           | 1255     | -                    | -   | 398 | mecA     |                          | LA           | CC398/LA              |
|          | Germany | Meat from cattle | Single, fresh, BCP | 4/74            | 2        | -                    | -   | -   | mecA     | 5                        | HA           | CC5/HA                |
|          |         |                  |                    | 1/74            | 8        | -                    | -   | -   | mecA     | 8                        | CA/HA        | CC8/CA/HA             |
|          |         | Meat from pigs   | Single, fresh, RM  | 1/380           | 693      | -                    | 9   | -   | mecA     | 1                        | LA           | CC1/LA                |
|          |         |                  |                    | 8/380           | 899      | -                    | 9   | -   | mecA     | 1                        | LA           | CC1/LA                |
|          |         |                  |                    | 1/380           | 20072    | -                    | 22  | -   | mecA     | 22                       | HA           | CC22/HA               |
|          |         |                  |                    | 1/380           | 8        | -                    | -   | -   | mecA     | 8                        | CA/HA        | CC8/CA/HA             |
|          |         |                  |                    | 10/380          | 11       | -                    | -   | -   | mecA     | 398                      | LA           | CC398/LA              |
|          |         |                  |                    | 7/380           | 34       | -                    | -   | -   | mecA     | 398                      | LA           | CC398/LA              |

| Category               | Country | Animal/food type  | Sample type/unit | No. of isolates | spa-type | PVL status/IEC genes | ST       | CC  | mec-gene | Inferred CC <sup>‡</sup> | LA, CA or HA | Inferred ST/CC & type |
|------------------------|---------|-------------------|------------------|-----------------|----------|----------------------|----------|-----|----------|--------------------------|--------------|-----------------------|
|                        |         |                   |                  | 1/380           | 127      | -                    | -        | -   | mecA     | 1                        | LA           | CC1/LA                |
|                        |         |                   |                  | 1/380           | 1419     | -                    | -        | -   | mecA     | 1                        | LA           | CC1/LA                |
|                        |         |                   |                  | 2/380           | 1430     | -                    | -        | -   | mecA     | 1                        | LA           | CC1/LA                |
|                        |         |                   |                  | 1/380           | 1451     | -                    | -        | -   | mecA     | 398                      | LA           | CC398/LA              |
|                        |         |                   |                  | 1/380           | 1580     | -                    | -        | -   | mecA     | 398                      | LA           | CC398/LA              |
|                        |         |                   |                  | 2/380           | 2011     | -                    | -        | -   | mecA     | 398                      | LA           | CC398/LA              |
|                        |         |                   |                  | 1/380           | 3512     | -                    | -        | -   | mecA     | 1                        | LA           | CC1/LA                |
| Food-producing animals | Belgium | Broilers          | Flock, OFM       | 1/96            | 11       | -                    | 398      | 398 | mecA     | -                        | LA           | CC398/LA              |
|                        |         | Fattening turkeys | Flock, OFM       | 3/27            | 11       | -                    | 398      | 398 | mecA     | -                        | LA           | CC398/LA              |
|                        |         |                   |                  | 2/27            | 11       | -                    | 873<br>1 | 398 | mecA     | -                        | LA           | CC398/LA              |
|                        | Germany | Breeding pigs     | Herd, OFM        | 1/351           | 3075     | -                    | 398      | 398 | mecA     | -                        | LA           | CC398/LA              |
|                        |         |                   |                  | 1/351           | 6608     | -                    | 398      | 398 | mecA     | -                        | LA           | CC398/LA              |
|                        |         |                   |                  | 1/351           | 16666    | -                    | 398      | 398 | mecA     | -                        | LA           | CC398/LA              |
|                        |         |                   |                  | 6/351           | 899      | -                    | 9        | -   | mecA     | 1                        | LA           | CC1/LA                |
|                        |         |                   |                  | 19/351          | 11       | -                    | -        | -   | mecA     | 398                      | LA           | CC398/LA              |
|                        |         |                   |                  | 38/351          | 34       | -                    | -        | -   | mecA     | 398                      | LA           | CC398/LA              |
|                        |         |                   |                  | 1/351           | 108      | -                    | -        | -   | mecA     | 398                      | LA           | CC398/LA              |
|                        |         |                   |                  | 2/351           | 1451     | -                    | -        | -   | mecA     | 398                      | LA           | CC398/LA              |
|                        |         |                   |                  | 1/351           | 1456     | -                    | -        | -   | mecA     | 398                      | LA           | CC398/LA              |
|                        |         |                   |                  | 6/351           | 2011     | -                    | -        | -   | mecA     | 398                      | LA           | CC398/LA              |
|                        |         |                   |                  | 2/351           | 2922     | -                    | -        | -   | mecA     | 398                      | LA           | CC398/LA              |
|                        |         |                   |                  | 1/351           | 4652     | -                    | -        | -   | mecA     | 398                      | LA           | CC398/LA              |
|                        |         |                   |                  | 3/351           | 6575     | -                    | -        | -   | mecA     | 398                      | LA           | CC398/LA              |

| Category | Country | Animal/food type | Sample type/unit | No. of isolates | spa-type | PVL status/IEC genes | ST | CC | mec-gene | Inferred CC <sup>‡</sup> | LA, CA or HA | Inferred ST/CC & type |
|----------|---------|------------------|------------------|-----------------|----------|----------------------|----|----|----------|--------------------------|--------------|-----------------------|
|          |         |                  |                  | 1/351           | 10204    | -                    | -  | -  | mecA     | 1                        | LA           | CC1/LA                |
|          |         |                  |                  | 1/351           | 19248    | -                    | -  | -  | mecA     | 398                      | LA           | CC398/LA              |

OFM, on-farm monitoring; RM, retail monitoring; SM, slaughterhouse monitoring; BCP, border control post; CC, clonal complex; ST, sequence type; CA, community-associated; HA, hospital-associated; LA, livestock-associated; IEC, immune evasion cluster; PVL, Pantón-Valentin leukocidin.

Details on the categorisation of *spa*-types can be found in the chapter on MRSA in the main part of the report (Section 6.3.3 Food and animals: Results of molecular typing of isolates). A summary of the information presented in this table is also visualised in Figure 60 of the MRSA chapter in the main part of the report.

\*Hybrid strain, t899 can be associated with CC1 and CC398.

<sup>‡</sup>For isolates where “inferred CC is stated, the CC is based on the *spa*-type or ST as reported in literature.

## E.6 Occurrence of resistance to selected antimicrobials in methicillin-resistant *Staphylococcus aureus*

**Table 1010: Occurrence of resistance (%) to selected antimicrobials in methicillin-resistant *Staphylococcus aureus* from food and animals, 2024.**

| Country                                          | N   | GEN  | KAN  | STR  | CHL  | RIF | CIP  | ERY  | CLI   | Q/D   | LZD | TIA   | MUP | FUS  | SMX | TMP   | TET   | VAN | CEF   | PEN   |
|--------------------------------------------------|-----|------|------|------|------|-----|------|------|-------|-------|-----|-------|-----|------|-----|-------|-------|-----|-------|-------|
| <b>Meat from turkey</b>                          |     |      |      |      |      |     |      |      |       |       |     |       |     |      |     |       |       |     |       |       |
| Germany                                          | 322 | 7.8  | 8.1  | 11.2 | 5.9  | 0.3 | 53.1 | 78.6 | 86.6  | 75.8  | 0.3 | 77.6  | 0.0 | 0.0  | 0.6 | 77.6  | 88.5  | 0   | 99.7  | 99.4  |
| Austria                                          | 28  | 14.3 | 17.9 | 17.9 | 7.1  | 0.0 | 60.7 | 71.4 | 75.0  | 64.3  | 3.6 | 67.9  | 0.0 | 0.0  | 0.0 | 53.6  | 92.9  | 0   | 100.0 | 100.0 |
| <b>Meat from broilers (<i>Gallus gallus</i>)</b> |     |      |      |      |      |     |      |      |       |       |     |       |     |      |     |       |       |     |       |       |
| Germany                                          | 124 | 4.0  | 1.6  | 19.4 | 0.0  | 0.0 | 16.9 | 71.8 | 93.5  | 88.7  | 0.0 | 91.1  | 0.8 | 0.8  | 0.8 | 83.1  | 95.2  | 0   | 100.0 | 100.0 |
| Austria                                          | 15  | 0.0  | 13.3 | 20.0 | 6.7  | 0.0 | 13.3 | 60.0 | 60.0  | 33.3  | 0.0 | 33.3  | 0.0 | 6.7  | 0.0 | 46.7  | 66.7  | 0   | 100.0 | 100.0 |
| <b>Crustaceans</b>                               |     |      |      |      |      |     |      |      |       |       |     |       |     |      |     |       |       |     |       |       |
| Germany                                          | 34  | 32.4 | 35.3 | 0.0  | 5.9  | 0.0 | 47.1 | 82.4 | 23.5  | 0.0   | 0.0 | 0.0   | 0.0 | 11.8 | 0.0 | 29.4  | 23.5  | 0   | 100.0 | 100.0 |
| <b><i>Gallus gallus</i> (fowl)</b>               |     |      |      |      |      |     |      |      |       |       |     |       |     |      |     |       |       |     |       |       |
| Germany                                          | 3   | 0.0  | 0.0  | 66.7 | 0.0  | 0.0 | 66.7 | 33.3 | 100.0 | 100.0 | 0.0 | 100.0 | 0.0 | 0.0  | 0.0 | 100.0 | 100.0 | 0   | 100.0 | 66.7  |
| <b>Small ruminants</b>                           |     |      |      |      |      |     |      |      |       |       |     |       |     |      |     |       |       |     |       |       |
| Portugal                                         | 8   | 0.0  | 0.0  | 0.0  | 25.0 | 0.0 | 12.5 | 0.0  | 12.5  | 0.0   | 0.0 | 12.5  | 0.0 | 0.0  | 0.0 | 0.0   | 62.5  | 0   | 100.0 | 37.5  |

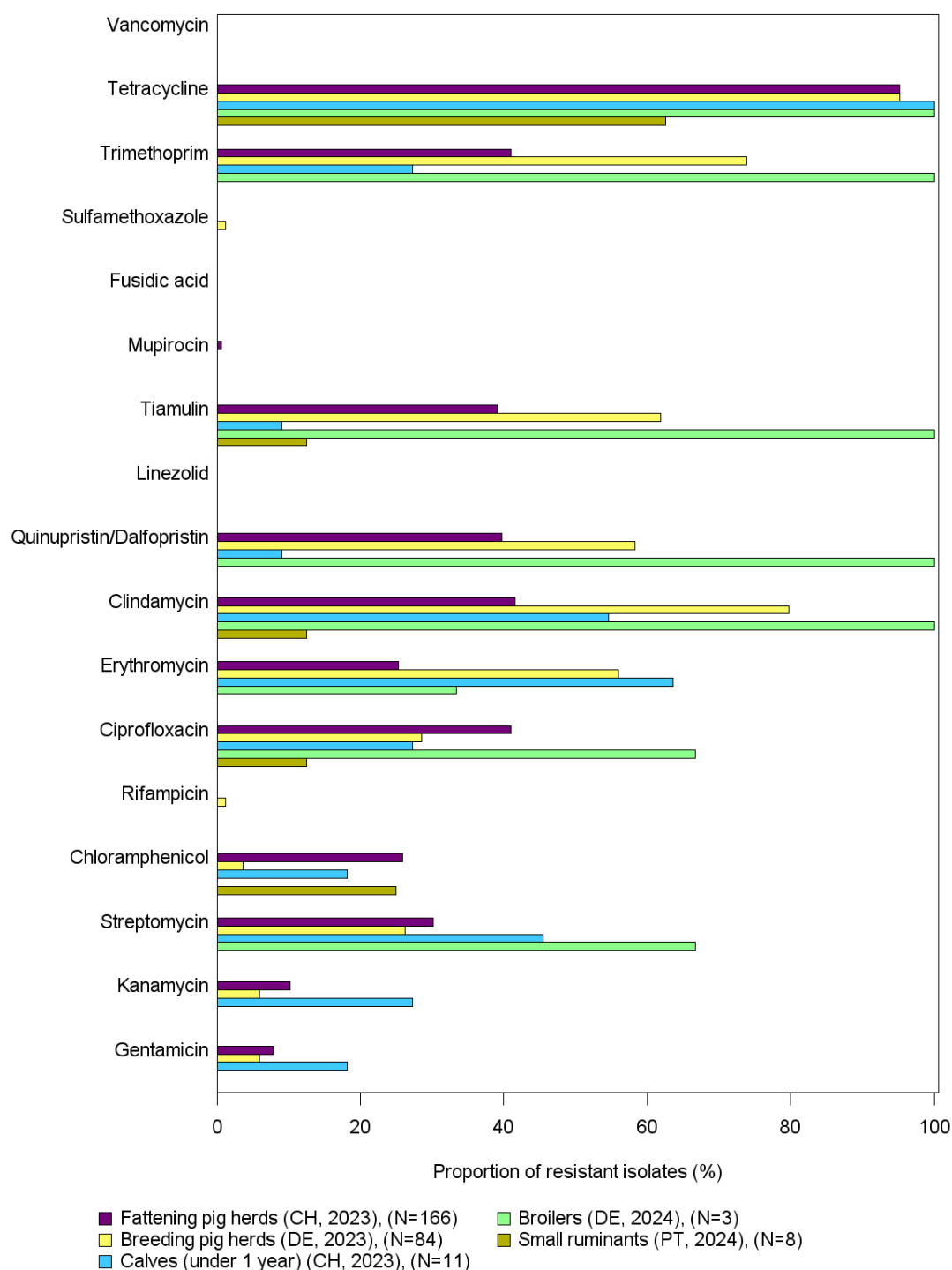
N, number of isolates tested; GEN, gentamicin; KAN, kanamycin; STR, streptomycin; CHL, chloramphenicol; RIF, rifampicin; CIP, ciprofloxacin; ERY, erythromycin; CLI, clindamycin; Q/D, quinupristin/dalfopristin; LZD, linezolid; TIA, tiamulin; MUP, mupirocin; FUS, fusidic acid; SMX, sulfamethoxazole; TMP, trimethoprim; TET, tetracycline; VAN, vancomycin; CEF: cefoxitin; PEN; Penicillin

**Table 1111: Occurrence of resistance (%) to selected antimicrobials in methicillin-resistant *Staphylococcus aureus* from food and animals, 2023.**

| Country                                      | N                | GEN  | KAN  | STR  | CHL  | RIF | CIP  | ERY  | CLI  | Q/D  | LZD | TIA  | MUP | FUS | SMX | TMP  | TET  | VAN |
|----------------------------------------------|------------------|------|------|------|------|-----|------|------|------|------|-----|------|-----|-----|-----|------|------|-----|
| <b>Cattle under 1 year</b>                   |                  |      |      |      |      |     |      |      |      |      |     |      |     |     |     |      |      |     |
| Switzerland                                  | 11               | 18.2 | 27.3 | 45.5 | 18.2 | 0   | 27.3 | 63.6 | 54.5 | 9.1  | 0   | 9.1  | 0   | 0   | 0   | 27.3 | 100  | 0   |
| <b>Pigs</b>                                  |                  |      |      |      |      |     |      |      |      |      |     |      |     |     |     |      |      |     |
| Switzerland                                  | 166              | 7.8  | 10.4 | 30.1 | 25.9 | 0   | 41.0 | 25.3 | 41.6 | 39.8 | 0   | 39.2 | 0.6 | 0   | 0   | 41.0 | 95.2 | 0   |
| Germany                                      | 84               | 6.0  | 6.0  | 26.2 | 3.6  | 1.2 | 28.6 | 56.0 | 79.8 | 58.3 | 0   | 61.9 | 0   | 0   | 1.2 | 73.8 | 95.2 | 0   |
| <b>Meat from pigs - fresh</b>                |                  |      |      |      |      |     |      |      |      |      |     |      |     |     |     |      |      |     |
| Austria                                      | 45               | 4.4  | 4.4  | 24.4 | 4.4  | 4.4 | 33.3 | 37.8 | 53.3 | 37.8 | 0   | 35.6 | 0   | 0   | 2.2 | 42.2 | 91.1 | 0   |
| Germany                                      | 37               | 16.2 | 10.8 | 16.2 | 8.1  | 0   | 40.5 | 59.5 | 56.8 | 27.0 | 0   | 27.0 | 0   | 0   | 0   | 54.1 | 70.3 | 0   |
| <b>Meat from cattle under 1 year - fresh</b> |                  |      |      |      |      |     |      |      |      |      |     |      |     |     |     |      |      |     |
| Austria                                      | 6 <sup>(d)</sup> | 0    | 0    | 0    | 0    | 0   | 33.3 | 50.0 | 100  | 66.7 | 0   | 66.7 | 0   | 0   | 0   | 88.3 | 66.7 | 0   |
| Germany                                      | 5 <sup>(e)</sup> | 80.0 | 80.0 | 0    | 0    | 0   | 20.0 | 20.0 | 20.0 | 0    | 0   | 20.0 | 0   | 0   | 0   | 0    | 20.0 | 0   |

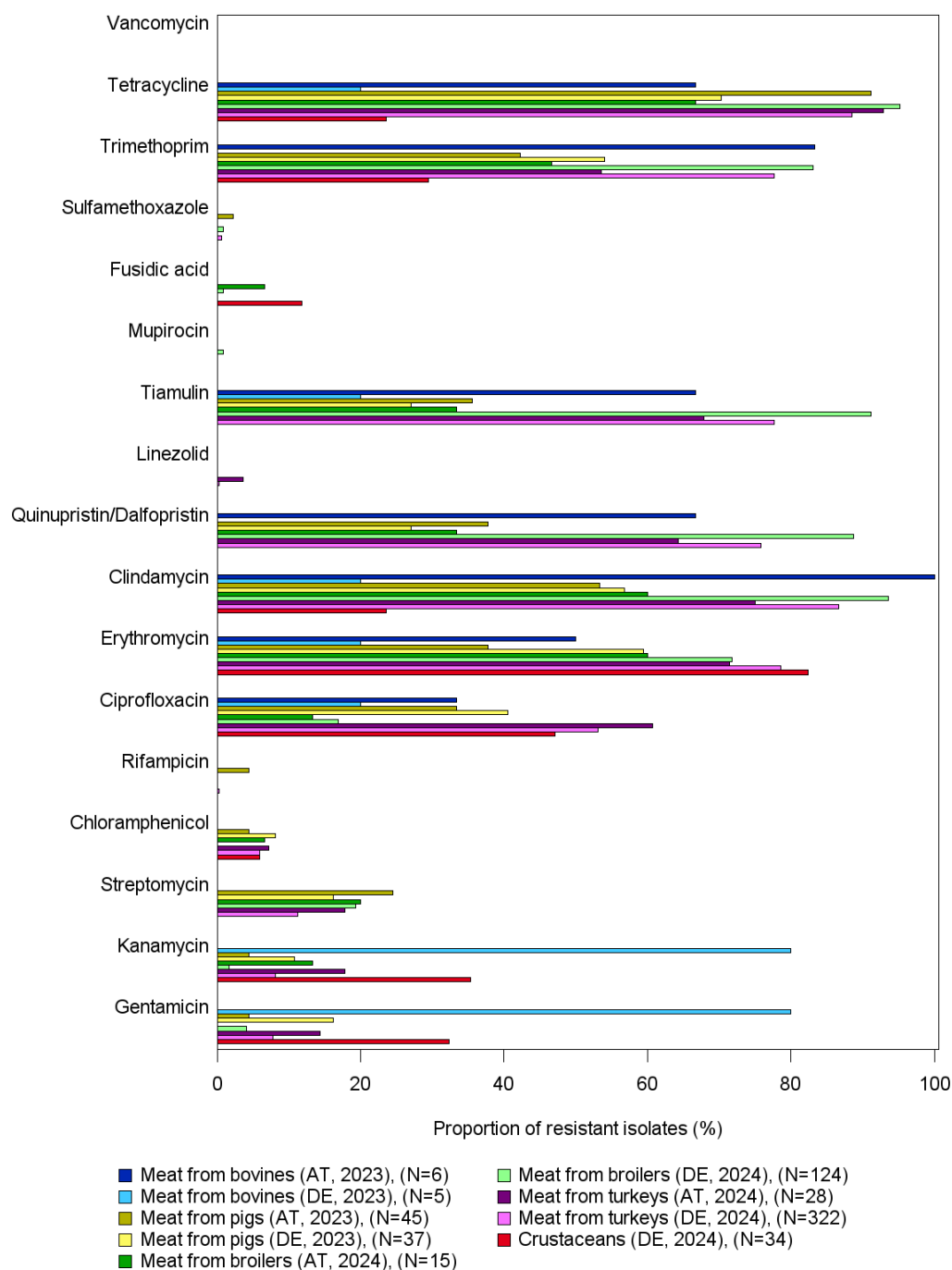
N, number of isolates tested; GEN, gentamicin; KAN, kanamycin; STR, streptomycin; CHL, chloramphenicol; RIF, rifampicin; CIP, ciprofloxacin; ERY, erythromycin; CLI, clindamycin; Q/D, quinupristin/dalfopristin; LZD, linezolid; TIA, tiamulin; MUP, mupirocin; FUS, fusidic acid; SMX, sulfamethoxazole; TMP, trimethoprim; TET, tetracycline; VAN, vancomycin. All MRSA isolates were resistant to penicillin and ceftiofur.

**Figure 1: Antimicrobial resistance in methicillin-resistant *Staphylococcus aureus* from targeted food-producing animals, 2023 and 2024.**



CH: Switzerland, DE: Germany. In 2023, all isolates were resistant to CEF and PEN. In 2024, six isolates (Germany and Portugal) were susceptible to penicillin.

**Figure 2: Antimicrobial resistance in methicillin-resistant *Staphylococcus aureus* from food, 2023 and 2024.**



AT: Austria, DE: Germany. In 2023, all isolates were resistant to penicillin and cefoxitin. In 2024, one German isolate was susceptible to cefoxitin and two to penicillin.



## E.7 MDR patterns in methicillin-resistant *Staphylococcus aureus*

**Table 1212: MDR patterns in methicillin-resistant *Staphylococcus aureus* isolates from food and animals, 2024.**

| MDR patterns in MRSA isolates       | Germany                  |                                                     |                       |                                   | Portugal              | Austria                 |                                                    |                       |
|-------------------------------------|--------------------------|-----------------------------------------------------|-----------------------|-----------------------------------|-----------------------|-------------------------|----------------------------------------------------|-----------------------|
|                                     | Meat from turkey (N=322) | Meat from broilers ( <i>Gallus gallus</i> ) (N=122) | Crustaceans (N=31)    | <i>Gallus gallus</i> (fowl) (N=3) | Small ruminants (N=3) | Meat from turkey (N=28) | Meat from broilers ( <i>Gallus gallus</i> ) (N=14) | Total                 |
|                                     | n (% of MDR isolates)    | n (% of MDR isolates)                               | n (% of MDR isolates) | n (% of MDR isolates)             | n (% of MDR isolates) | n (% of MDR isolates)   | n (% of MDR isolates)                              | n (% of MDR isolates) |
| CEF-CLI-ERY-PEN-Q/D-TET-TIA-TMP     | 90 (28)                  | 61 (50)                                             |                       | 1 (33.3)                          |                       | 4 (14.3)                | 2 (14.3)                                           | 158 (30.2)            |
| CEF-CIP-CLI-ERY-PEN-Q/D-TET-TIA-TMP | 74 (23)                  | 3 (2.5)                                             |                       |                                   |                       | 5 (17.9)                |                                                    | 82 (15.7)             |
| CEF-CLI-PEN-Q/D-TET-TIA-TMP         | 11 (3.4)                 | 10 (8.2)                                            |                       |                                   |                       |                         | 1 (7.1)                                            | 22 (4.2)              |
| CEF-CIP-PEN                         | 17 (5.3)                 | 1 (0.8)                                             | 2 (6.5)               |                                   |                       |                         |                                                    | 20 (3.8)              |
| CEF-CLI-ERY-PEN-Q/D-TET-TIA         | 5 (1.6)                  | 10 (8.2)                                            |                       |                                   |                       |                         |                                                    | 15 (2.9)              |
| CEF-CIP-CLI-ERY-GEN-KAN-PEN-TET-TMP | 12 (3.7)                 |                                                     |                       |                                   |                       | 2 (7.1)                 |                                                    | 14 (2.7)              |
| CEF-CIP-CLI-PEN-Q/D-STR-TET-TIA-TMP | 1 (0.3)                  | 12 (9.8)                                            |                       | 1 (33.3)                          |                       |                         |                                                    | 14 (2.7)              |

|                                             |         |         |          |  |          |          |          |          |
|---------------------------------------------|---------|---------|----------|--|----------|----------|----------|----------|
| CEF-CLI-ERY-PEN-Q/D-STR-TET-TIA-TMP         | 8 (2.5) | 5 (4.1) |          |  |          |          |          | 13 (2.5) |
| CEF-CIP-CLI-ERY-PEN-Q/D-TET-TIA             | 5 (1.6) | 1 (0.8) |          |  |          | 6 (21.4) |          | 12 (2.3) |
| CEF-ERY-PEN                                 |         |         | 7 (22.6) |  |          |          | 2 (14.3) | 9 (1.7)  |
| CEF-PEN-TET                                 | 3 (0.9) | 3 (2.5) |          |  | 1 (33.3) | 2 (7.1)  |          | 9 (1.7)  |
| CEF-CIP-PEN-TMP                             | 8 (2.5) |         |          |  |          |          |          | 8 (1.5)  |
| CEF-CIP-CLI-ERY-PEN-Q/D-STR-TET-TIA-TMP     | 6 (1.9) | 1 (0.8) |          |  |          |          |          | 7 (1.3)  |
| CEF-CIP-CLI-PEN-Q/D-TET-TIA-TMP             | 7 (2.2) |         |          |  |          |          |          | 7 (1.3)  |
| CEF-CLI-PEN-Q/D-TET-TIA                     | 4 (1.2) | 2 (1.6) |          |  |          |          | 1 (7.1)  | 7 (1.3)  |
| CEF-CLI-PEN-Q/D-STR-TET-TIA-TMP             | 2 (0.6) | 2 (1.6) |          |  |          | 1 (3.6)  | 1 (7.1)  | 6 (1.1)  |
| CEF-CHL-PEN                                 | 4 (1.2) |         |          |  |          | 1 (3.6)  |          | 5 (1)    |
| CEF-CIP-CLI-ERY-GEN-KAN-PEN-Q/D-STR-TET-TIA | 4 (1.2) |         |          |  |          | 1 (3.6)  |          | 5 (1)    |
| CEF-CIP-CLI-ERY-PEN-TET-TMP                 | 2 (0.6) | 2 (1.6) |          |  |          |          | 1 (7.1)  | 5 (1)    |
| CEF-CLI-ERY-PEN-TET                         | 1 (0.3) |         | 3 (9.7)  |  |          | 1 (3.6)  |          | 5 (1)    |
| CEF-CIP-CLI-ERY-PEN-STR-TET                 | 4 (1.2) |         |          |  |          |          |          | 4 (0.8)  |
| CEF-CIP-ERY-FUS-GEN-KAN-PEN                 |         |         | 4 (12.9) |  |          |          |          | 4 (0.8)  |
| CEF-CIP-ERY-PEN                             |         |         | 4 (12.9) |  |          |          |          | 4 (0.8)  |
| CEF-CIP-PEN-TET-TIA-TMP                     | 3 (0.9) |         |          |  |          | 1 (3.6)  |          | 4 (0.8)  |
| CEF-CHL-CIP-CLI-ERY-PEN-Q/D-STR-TET-TIA-TMP | 3 (0.9) |         |          |  |          |          |          | 3 (0.6)  |
| CEF-CLI-ERY-GEN-PEN-Q/D-                    |         | 3 (2.5) |          |  |          |          |          | 3 (0.6)  |

|                                         |         |  |         |  |  |         |         |         |
|-----------------------------------------|---------|--|---------|--|--|---------|---------|---------|
| STR-TET-TIA-TMP                         |         |  |         |  |  |         |         |         |
| CEF-CLI-ERY-KAN-PEN-Q/D-TET-TIA-TMP     | 3 (0.9) |  |         |  |  |         |         | 3 (0.6) |
| CEF-CHL-CIP-CLI-ERY-GEN-KAN-PEN-TET-TMP |         |  | 2 (6.5) |  |  |         |         | 2 (0.4) |
| CEF-CHL-CIP-CLI-ERY-PEN-Q/D-TET-TIA-TMP | 2 (0.6) |  |         |  |  |         |         | 2 (0.4) |
| CEF-CHL-CLI-ERY-PEN-TET                 | 2 (0.6) |  |         |  |  |         |         | 2 (0.4) |
| CEF-CIP-CLI-ERY-GEN-KAN-PEN-TMP         |         |  | 2 (6.5) |  |  |         |         | 2 (0.4) |
| CEF-CIP-CLI-ERY-PEN-Q/D-STR-TET-TIA     | 2 (0.6) |  |         |  |  |         |         | 2 (0.4) |
| CEF-CIP-CLI-ERY-PEN-Q/D-TET-TMP         | 2 (0.6) |  |         |  |  |         |         | 2 (0.4) |
| CEF-CIP-CLI-ERY-PEN-TET                 | 1 (0.3) |  |         |  |  |         | 1 (7.1) | 2 (0.4) |
| CEF-CIP-CLI-ERY-PEN-TET-TIA-TMP         | 2 (0.6) |  |         |  |  |         |         | 2 (0.4) |
| CEF-CIP-PEN-TET                         | 2 (0.6) |  |         |  |  |         |         | 2 (0.4) |
| CEF-CLI-ERY-GEN-KAN-PEN-Q/D-TET-TIA-TMP | 2 (0.6) |  |         |  |  |         |         | 2 (0.4) |
| CEF-CLI-ERY-GEN-PEN-Q/D-TET-TIA-TMP     | 2 (0.6) |  |         |  |  |         |         | 2 (0.4) |
| CEF-CLI-ERY-GEN-PEN-TET                 | 2 (0.6) |  |         |  |  |         |         | 2 (0.4) |
| CEF-CLI-ERY-PEN-Q/D-STR-TET-TIA         | 2 (0.6) |  |         |  |  |         |         | 2 (0.4) |
| CEF-CLI-ERY-PEN-Q/D-TET                 | 2 (0.6) |  |         |  |  |         |         | 2 (0.4) |
| CEF-ERY-KAN-PEN-STR                     | 1 (0.3) |  |         |  |  | 1 (3.6) |         | 2 (0.4) |
| CEF-ERY-KAN-PEN-STR-TET                 | 1 (0.3) |  |         |  |  |         | 1 (7.1) | 2 (0.4) |
| CEF-PEN-TMP                             |         |  | 1 (3.2) |  |  |         | 1 (7.1) | 2 (0.4) |

|                                                 |         |  |  |  |          |         |         |         |
|-------------------------------------------------|---------|--|--|--|----------|---------|---------|---------|
| CEF-CHL-CIP-CLI-ERY-KAN-PEN-Q/D-STR-TET-TIA-TMP | 1 (0.3) |  |  |  |          |         |         | 1 (0.2) |
| CEF-CHL-CIP-CLI-LZD-PEN-Q/D-STR-TET-TIA-TMP     |         |  |  |  |          | 1 (3.6) |         | 1 (0.2) |
| CEF-CHL-CIP-CLI-LZD-PEN-Q/D-TIA                 | 1 (0.3) |  |  |  |          |         |         | 1 (0.2) |
| CEF-CHL-CIP-CLI-PEN-Q/D-TIA                     | 1 (0.3) |  |  |  |          |         |         | 1 (0.2) |
| CEF-CHL-CIP-CLI-PEN-TET-TIA                     |         |  |  |  | 1 (33.3) |         |         | 1 (0.2) |
| CEF-CHL-CIP-CLI-PEN-TIA                         | 1 (0.3) |  |  |  |          |         |         | 1 (0.2) |
| CEF-CHL-CIP-PEN                                 | 1 (0.3) |  |  |  |          |         |         | 1 (0.2) |
| CEF-CHL-CIP-PEN-STR                             | 1 (0.3) |  |  |  |          |         |         | 1 (0.2) |
| CEF-CHL-CLI-ERY-GEN-KAN-PEN-TET-TMP             | 1 (0.3) |  |  |  |          |         |         | 1 (0.2) |
| CEF-CHL-CLI-ERY-PEN-Q/D-TET-TIA-TMP             | 1 (0.3) |  |  |  |          |         |         | 1 (0.2) |
| CEF-CHL-CLI-ERY-PEN-STR-TET                     |         |  |  |  |          |         | 1 (7.1) | 1 (0.2) |
| CEF-CHL-TET                                     |         |  |  |  | 1 (33.3) |         |         | 1 (0.2) |
| CEF-CIP-CLI-ERY-GEN-KAN-PEN-RIF-TET-TMP         | 1 (0.3) |  |  |  |          |         |         | 1 (0.2) |
| CEF-CIP-CLI-ERY-PEN                             | 1 (0.3) |  |  |  |          |         |         | 1 (0.2) |
| CEF-CIP-CLI-ERY-PEN-Q/D-SMX-TET-TIA-TMP         | 1 (0.3) |  |  |  |          |         |         | 1 (0.2) |
| CEF-CIP-CLI-ERY-PEN-SMX-TET-TMP                 | 1 (0.3) |  |  |  |          |         |         | 1 (0.2) |
| CEF-CIP-CLI-ERY-Q/D-TET-TIA-TMP                 | 1 (0.3) |  |  |  |          |         |         | 1 (0.2) |

|                                     |         |         |         |          |  |         |         |         |
|-------------------------------------|---------|---------|---------|----------|--|---------|---------|---------|
| CEF-CIP-CLI-ERY-TET-TMP             | 1 (0.3) |         |         |          |  |         |         | 1 (0.2) |
| CEF-CIP-CLI-PEN-TIA                 | 1 (0.3) |         |         |          |  |         |         | 1 (0.2) |
| CEF-CIP-CLI-Q/D-STR-TET-TIA-TMP     |         |         |         | 1 (33.3) |  |         |         | 1 (0.2) |
| CEF-CIP-ERY-GEN-KAN-PEN-TMP         |         |         | 1 (3.2) |          |  |         |         | 1 (0.2) |
| CEF-CIP-ERY-PEN-TMP                 |         |         | 1 (3.2) |          |  |         |         | 1 (0.2) |
| CEF-CIP-GEN-KAN-PEN-TET-TMP         |         |         |         |          |  | 1 (3.6) |         | 1 (0.2) |
| CEF-CIP-PEN-SMX-TET-TIA-TMP         |         | 1 (0.8) |         |          |  |         |         | 1 (0.2) |
| CEF-CLI-ERY-GEN-KAN-PEN             |         | 1 (0.8) |         |          |  |         |         | 1 (0.2) |
| CEF-CLI-ERY-GEN-KAN-PEN-STR-TET-TMP |         | 1 (0.8) |         |          |  |         |         | 1 (0.2) |
| CEF-CLI-ERY-GEN-KAN-PEN-TET-TMP     |         |         | 1 (3.2) |          |  |         |         | 1 (0.2) |
| CEF-CLI-ERY-GEN-PEN-TET-TMP         | 1 (0.3) |         |         |          |  |         |         | 1 (0.2) |
| CEF-CLI-ERY-KAN-PEN-TET-TMP         |         |         |         |          |  |         | 1 (7.1) | 1 (0.2) |
| CEF-CLI-ERY-PEN-TET-TIA             | 1 (0.3) |         |         |          |  |         |         | 1 (0.2) |
| CEF-CLI-PEN-TET-TIA-TMP             |         | 1 (0.8) |         |          |  |         |         | 1 (0.2) |
| CEF-CLI-PEN-TIA-TMP                 |         | 1 (0.8) |         |          |  |         |         | 1 (0.2) |
| CEF-ERY-FUS-MUP-PEN                 |         | 1 (0.8) |         |          |  |         |         | 1 (0.2) |
| CEF-ERY-GEN-KAN-PEN                 |         |         | 1 (3.2) |          |  |         |         | 1 (0.2) |
| CEF-ERY-KAN-PEN-TET-TMP             |         |         | 1 (3.2) |          |  |         |         | 1 (0.2) |
| CEF-ERY-PEN-TET-TIA-TMP             | 1 (0.3) |         |         |          |  |         |         | 1 (0.2) |
| CEF-ERY-PEN-TET-TMP                 |         |         | 1 (3.2) |          |  |         |         | 1 (0.2) |
| CEF-FUS-PEN                         |         |         |         |          |  |         | 1 (7.1) | 1 (0.2) |

|                             |         |  |  |  |  |         |  |         |
|-----------------------------|---------|--|--|--|--|---------|--|---------|
| CEF-PEN-STR-TET             |         |  |  |  |  | 1 (3.6) |  | 1 (0.2) |
| CEF-PEN-TET-TIA             | 1 (0.3) |  |  |  |  |         |  | 1 (0.2) |
| CIP-CLI-ERY-PEN-Q/D-TET-TIA | 1 (0.3) |  |  |  |  |         |  | 1 (0.2) |

MDR, multidrug resistant; N, total number of isolates tested; n, number of isolates with MDR pattern; CEF, ceftiofur; PNC, penicillin; GEN, gentamicin; KAN, kanamycin; STR, streptomycin; CHL, chloramphenicol; RIF, rifampicin; CIP, ciprofloxacin; ERY, erythromycin; CLI, clindamycin; Q/D, quinupristin/dalfopristin; TIA, tiamulin; SMX, sulfamethoxazole; TMP, trimethoprim; TET, tetracycline.

**Table 1313: MDR patterns in methicillin-resistant *Staphylococcus aureus* isolates from food and animals, 2023.**

| MDR patterns in MRSA isolates       | Germany                     |                                            |                             | Switzerland                             |                             | Austria                     |                                            |                             |
|-------------------------------------|-----------------------------|--------------------------------------------|-----------------------------|-----------------------------------------|-----------------------------|-----------------------------|--------------------------------------------|-----------------------------|
|                                     | Pigs<br>(N=84)              | Meat<br>from<br>bovine<br>animals<br>(N=4) | Meat<br>from pig<br>(N=37)  | Cattle<br>(bovine<br>animals)<br>(N=11) | Pigs<br>(N=166)             | Meat<br>from pig<br>(N=45)  | Meat<br>from<br>bovine<br>animals<br>(N=6) | Total                       |
|                                     | n (% of<br>MDR<br>isolates) | n (% of<br>MDR<br>isolates)                | n (% of<br>MDR<br>isolates) | n (% of<br>MDR<br>isolates)             | n (% of<br>MDR<br>isolates) | n (% of<br>MDR<br>isolates) | n (% of<br>MDR<br>isolates)                | n (% of<br>MDR<br>isolates) |
| CEF-CLI-ERY-PEN-Q/D-TET-TIA-TMP     | 16 (19)                     |                                            | 5 (13.5)                    | 1 (9.1)                                 | 19 (11.4)                   | 2 (4.4)                     |                                            | 43 (12.2)                   |
| CEF-PEN-TET                         | 6 (7.1)                     |                                            | 2 (5.4)                     | 1 (9.1)                                 | 14 (8.4)                    | 9 (20)                      |                                            | 32 (9.1)                    |
| CEF-CHL-CIP-PEN-TET                 |                             |                                            |                             |                                         | 31 (18.7)                   |                             |                                            | 31 (8.8)                    |
| CEF-CLI-PEN-Q/D-TET-TIA-TMP         | 10 (11.9)                   |                                            |                             |                                         | 10 (6)                      | 2 (4.4)                     | 1 (16.7)                                   | 23 (6.5)                    |
| CEF-CLI-ERY-PEN-Q/D-STR-TET-TIA-TMP | 4 (4.8)                     |                                            | 2 (5.4)                     |                                         | 14 (8.4)                    |                             |                                            | 20 (5.7)                    |
| CEF-PEN-STR-TET                     | 1 (1.2)                     |                                            |                             | 1 (9.1)                                 | 14 (8.4)                    | 2 (4.4)                     |                                            | 18 (5.1)                    |
| CEF-CIP-CLI-ERY-PEN-TET-TMP         | 5 (6)                       |                                            | 4 (10.8)                    | 1 (9.1)                                 |                             |                             | 1 (16.7)                                   | 11 (3.1)                    |
| CEF-CIP-PEN                         |                             |                                            | 7 (18.9)                    |                                         |                             | 4 (8.9)                     |                                            | 11 (3.1)                    |

|                                         |         |  |         |         |         |         |          |          |
|-----------------------------------------|---------|--|---------|---------|---------|---------|----------|----------|
| CEF-CIP-PEN-TET                         | 1 (1.2) |  | 1 (2.7) |         | 7 (4.2) | 2 (4.4) |          | 11 (3.1) |
| CEF-CLI-PEN-Q/D-TIA-TMP                 |         |  |         |         | 8 (4.8) |         | 2 (33.3) | 10 (2.8) |
| CEF-CLI-PEN-Q/D-STR-TET-TIA-TMP         | 5 (6)   |  |         |         | 1 (0.6) | 3 (6.7) |          | 9 (2.5)  |
| CEF-CIP-CLI-PEN-Q/D-STR-TET-TIA-TMP     | 2 (2.4) |  |         |         | 4 (2.4) | 1 (2.2) |          | 7 (2)    |
| CEF-CIP-PEN-STR-TET                     |         |  |         |         | 7 (4.2) |         |          | 7 (2)    |
| CEF-GEN-KAN-PEN-TET                     |         |  |         | 1 (9.1) | 6 (3.6) |         |          | 7 (2)    |
| CEF-CIP-CLI-ERY-PEN-Q/D-TET-TIA-TMP     | 2 (2.4) |  |         |         | 1 (0.6) | 2 (4.4) | 1 (16.7) | 6 (1.7)  |
| CEF-PEN-TET-TMP                         |         |  | 2 (5.4) |         | 4 (2.4) |         |          | 6 (1.7)  |
| CEF-CIP-CLI-ERY-PEN-Q/D-STR-TET-TIA-TMP | 2 (2.4) |  | 1 (2.7) |         | 1 (0.6) | 1 (2.2) |          | 5 (1.4)  |
| CEF-CLI-ERY-PEN-Q/D-TET-TIA             | 3 (3.6) |  | 1 (2.7) |         |         | 1 (2.2) |          | 5 (1.4)  |
| CEF-CLI-ERY-GEN-KAN-PEN-TET             | 1 (1.2) |  |         |         | 3 (1.8) |         |          | 4 (1.1)  |
| CEF-CLI-ERY-PEN-TET                     | 1 (1.2) |  |         | 1 (9.1) |         | 1 (2.2) | 1 (16.7) | 4 (1.1)  |
| CEF-CHL-CIP-CLI-KAN-PEN-Q/D-STR-TET-TIA |         |  |         |         | 3 (1.8) |         |          | 3 (0.8)  |
| CEF-CHL-CIP-PEN-STR-TET                 |         |  |         |         | 3 (1.8) |         |          | 3 (0.8)  |



|                                     |         |        |         |          |         |         |  |         |
|-------------------------------------|---------|--------|---------|----------|---------|---------|--|---------|
| CEF-CIP-PEN-TET-TMP                 |         |        |         |          | 2 (1.2) | 1 (2.2) |  | 3 (0.8) |
| CEF-CLI-ERY-GEN-KAN-PEN-TET-TMP     | 1 (1.2) |        | 2 (5.4) |          |         |         |  | 3 (0.8) |
| CEF-CLI-ERY-GEN-PEN-TET-TMP         | 1 (1.2) |        | 1 (2.7) |          |         | 1 (2.2) |  | 3 (0.8) |
| CEF-GEN-KAN-PEN                     |         | 3 (75) |         |          |         |         |  | 3 (0.8) |
| CEF-GEN-KAN-PEN-TET-TMP             | 1 (1.2) |        |         | 1 (9.1)  | 1 (0.6) |         |  | 3 (0.8) |
| CEF-CHL-CIP-CLI-ERY-PEN-STR-TET     |         |        |         | 2 (18.2) |         |         |  | 2 (0.6) |
| CEF-CHL-CIP-CLI-ERY-PEN-TET         |         |        |         |          | 2 (1.2) |         |  | 2 (0.6) |
| CEF-CHL-CLI-ERY-PEN-TET-TMP         | 1 (1.2) |        |         |          |         | 1 (2.2) |  | 2 (0.6) |
| CEF-CHL-PEN-TET                     |         |        | 1 (2.7) |          |         | 1 (2.2) |  | 2 (0.6) |
| CEF-CIP-CLI-ERY-PEN                 | 1 (1.2) |        | 1 (2.7) |          |         |         |  | 2 (0.6) |
| CEF-CIP-CLI-ERY-PEN-STR-TET         |         |        |         |          |         | 2 (4.4) |  | 2 (0.6) |
| CEF-CIP-CLI-PEN-Q/D-TET-TIA-TMP     | 2 (2.4) |        |         |          |         |         |  | 2 (0.6) |
| CEF-CLI-ERY-GEN-KAN-PEN-STR-TET-TMP | 1 (1.2) |        |         |          |         | 1 (2.2) |  | 2 (0.6) |
| CEF-CLI-ERY-PEN-Q/D-TET-TMP         | 1 (1.2) |        |         |          |         | 1 (2.2) |  | 2 (0.6) |
| CEF-CLI-ERY-PEN-STR-TET             | 1 (1.2) |        |         | 1 (9.1)  |         |         |  | 2 (0.6) |

|                                                 |         |        |         |  |         |         |  |         |
|-------------------------------------------------|---------|--------|---------|--|---------|---------|--|---------|
| CEF-CLI-ERY-PEN-TET-TMP                         | 1 (1.2) |        |         |  |         | 1 (2.2) |  | 2 (0.6) |
| CEF-PEN-TET-TIA                                 | 2 (2.4) |        |         |  |         |         |  | 2 (0.6) |
| CEF-CHL-CIP-CLI-ERY-GEN-KAN-PEN-Q/D-STR-TET-TIA |         |        |         |  | 1 (0.6) |         |  | 1 (0.3) |
| CEF-CHL-CIP-KAN-PEN-TET                         |         |        |         |  | 1 (0.6) |         |  | 1 (0.3) |
| CEF-CHL-CIP-MUP-PEN-TET                         |         |        |         |  | 1 (0.6) |         |  | 1 (0.3) |
| CEF-CHL-CIP-PEN-STR-TET-TMP                     | 1 (1.2) |        |         |  |         |         |  | 1 (0.3) |
| CEF-CHL-CIP-PEN-TMP                             | 1 (1.2) |        |         |  |         |         |  | 1 (0.3) |
| CEF-CHL-CLI-ERY-GEN-PEN-TET-TMP                 |         |        | 1 (2.7) |  |         |         |  | 1 (0.3) |
| CEF-CHL-CLI-ERY-KAN-PEN-Q/D-STR-TET-TIA         |         |        | 1 (2.7) |  |         |         |  | 1 (0.3) |
| CEF-CHL-PEN-STR-TET                             |         |        |         |  | 1 (0.6) |         |  | 1 (0.3) |
| CEF-CIP-CLI-ERY-GEN-KAN-PEN-TET-TIA             |         | 1 (25) |         |  |         |         |  | 1 (0.3) |
| CEF-CIP-CLI-ERY-KAN-PEN-Q/D-STR-TET-TIA-TMP     |         |        |         |  |         | 1 (2.2) |  | 1 (0.3) |
| CEF-CIP-CLI-ERY-PEN-Q/D-RIF-TET-TIA             | 1 (1.2) |        |         |  |         |         |  | 1 (0.3) |

|                                         |         |  |         |  |         |         |  |         |
|-----------------------------------------|---------|--|---------|--|---------|---------|--|---------|
| CEF-CIP-CLI-ERY-PEN-Q/D-TET-TIA         |         |  |         |  |         | 1 (2.2) |  | 1 (0.3) |
| CEF-CIP-CLI-ERY-PEN-STR                 | 1 (1.2) |  |         |  |         |         |  | 1 (0.3) |
| CEF-CIP-CLI-ERY-PEN-STR-TET-TMP         | 1 (1.2) |  |         |  |         |         |  | 1 (0.3) |
| CEF-CIP-CLI-ERY-PEN-TET                 | 1 (1.2) |  |         |  |         |         |  | 1 (0.3) |
| CEF-CIP-CLI-KAN-PEN-Q/D-STR-TET-TIA-TMP |         |  |         |  | 1 (0.6) |         |  | 1 (0.3) |
| CEF-CIP-CLI-PEN-TET-TMP                 |         |  |         |  | 1 (0.6) |         |  | 1 (0.3) |
| CEF-CIP-GEN-PEN-TET                     |         |  |         |  | 1 (0.6) |         |  | 1 (0.3) |
| CEF-CIP-KAN-PEN-STR-TET-TMP             | 1 (1.2) |  |         |  |         |         |  | 1 (0.3) |
| CEF-CIP-PEN-Q/D-TET                     |         |  |         |  | 1 (0.6) |         |  | 1 (0.3) |
| CEF-CIP-PEN-SMX                         | 1 (1.2) |  |         |  |         |         |  | 1 (0.3) |
| CEF-CIP-PEN-STR                         |         |  | 1 (2.7) |  |         |         |  | 1 (0.3) |
| CEF-CIP-PEN-STR-TET-TMP                 | 1 (1.2) |  |         |  |         |         |  | 1 (0.3) |
| CEF-CLI-ERY-GEN-PEN-STR-TET-TMP         |         |  | 1 (2.7) |  |         |         |  | 1 (0.3) |
| CEF-CLI-ERY-GEN-PEN-TET                 |         |  | 1 (2.7) |  |         |         |  | 1 (0.3) |

|                                     |         |  |         |         |         |         |  |         |
|-------------------------------------|---------|--|---------|---------|---------|---------|--|---------|
| CEF-CLI-ERY-PEN-Q/D-RIF-TET-TIA-TMP |         |  |         |         |         | 1 (2.2) |  | 1 (0.3) |
| CEF-CLI-ERY-PEN-Q/D-STR-TET-TIA     | 1 (1.2) |  |         |         |         |         |  | 1 (0.3) |
| CEF-CLI-PEN-Q/D-TET-TIA             |         |  |         |         |         | 1 (2.2) |  | 1 (0.3) |
| CEF-CLI-PEN-TET-TIA-TMP             | 1 (1.2) |  |         |         |         |         |  | 1 (0.3) |
| CEF-ERY-KAN-PEN                     |         |  | 1 (2.7) |         |         |         |  | 1 (0.3) |
| CEF-ERY-KAN-PEN-STR-TET             |         |  |         | 1 (9.1) |         |         |  | 1 (0.3) |
| CEF-ERY-PEN-TET-TIA-TMP             |         |  |         |         | 1 (0.6) |         |  | 1 (0.3) |
| CEF-GEN-KAN-PEN-Q/D-TET-TIA         |         |  |         |         | 1 (0.6) |         |  | 1 (0.3) |
| CEF-PEN-Q/D-TET                     |         |  |         |         | 1 (0.6) |         |  | 1 (0.3) |
| CEF-PEN-RIF-TET                     |         |  |         |         |         | 1 (2.2) |  | 1 (0.3) |
| CEF-PEN-SMX-TET                     |         |  |         |         |         | 1 (2.2) |  | 1 (0.3) |
| CEF-PEN-TET-TIA-TMP                 | 1 (1.2) |  |         |         |         |         |  | 1 (0.3) |
| CEF-PEN-TMP                         |         |  | 1 (2.7) |         |         |         |  | 1 (0.3) |

MDR, multidrug resistant; N, total number of isolates tested; n, number of isolates with MDR pattern; CEF, cefoxitin; PNC, penicillin; GEN, gentamicin; KAN, kanamycin; STR, streptomycin; CHL, chloramphenicol; RIF, rifampicin; CIP, ciprofloxacin; ERY, erythromycin; CLI, clindamycin; Q/D, quinupristin/dalfopristin; TIA, tiamulin; SMX, sulfamethoxazole; TMP, trimethoprim; TET, tetracycline.